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SUPERSEDING
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(See 6.9)

MILITARY SPECIFICATION

BATTERY CELLS AND ELEMENTS, LEAD-ACID,

LEAD-ANTIMONY, MAIN STORAGE, SUBMARINE;

GENERAL SPECIFICATION FOR (METRIC)

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers lead-acid, lead-antimony main storage battery cells and elements for submarines.

1.2 Classification. Battery cells and elements shall be of the types specified in 1.2.1 and the classes specified in 1.2.2, as specified (see 6.2.1 and 6.7).

1.2.1 Type. Type shall be determined by the electrical capacity and configuration of the cell. Types shall be as follows:

Type SARGO I
Type SARGO II
Type GUPPY I
Type GUPPY I, MOD A

1.2.2 Classification. Class shall be determined by the number of cells and battery groups in a complete battery installation. Classes shall be as follows:

Class 2 - 252 cells in two 126-cell groups.
Class 3 - 504 cells in two 252-cell groups.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL
O-S-801 - Sulfuric Acid, Electrolyte; for Storage Batteries.
FF-W-92 - Washers, Metal, Flat (Plain).

MILITARY
MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements For.
MIL-C-915 - Cable and Cord, Electrical, For Shipboard Use; General Specification For.
MIL-C-915/6 - Cable Electrical, 600 Volts, Types SHOF, DHOF, THOF, and FHOF.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Ship Engineering Center, SEC 6124, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6140

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MILITARY (Cont'd)
MIL-S-1222 - Studs, Bolts, Hex Cap Screws, and Nuts.
MIL-C-2687 - Compound, Battery-Sealing.
MIL-M-15071 - Manuals, Technical, Equipment and Systems Content Requirements For.
DOD-P-15816 - Parts, Hard Rubber for Use with Submarine Storage Batteries.
DOD-P-15817 - Parts, Synthetic Rubber, for Use with Batteries or in Battery Compartments of Submarines.
MIL-J-16128 - Jars, Battery, Laminated, Rubber, Submarine.
MIL-A-22681 - Arrester, Flash, Main Storage Battery (Submarine Use), General Specification For.
MIL-N-25027 - Nut, Self-Locking, 250°F, 450°F, and 800°F, 125 KSI FTU, 60 KSI FTU, and 30 KSI FTU.
MIL-I-45208 - Inspection System Requirements.

STANDARDS

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MS 21045 - Nut, Self-Locking, Hexagon-Regular Height, 450°F, 125 KSI FTU.

DRAWINGS

MILITARY
NAVSHIPS 9-S-5091-L - Submarine Storage Battery, Curves and Data.
NAVSHIPS S6204-H-3,211,595 - Submarine Main Storage Battery Cell Jumpers.

PUBLICATIONS

MILITARY
NAVSEA S9086-GI-STM-010/CH-223 V1 - Naval Ships Technical Manual, Chapter 223, Lead-Acid Submarine Storage Batteries.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting agency or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
A276 - Stainless and Heat-Resisting Steel Bars and Shapes, Specification for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Ratings, Rules and Regulations.

(Application for copies should be addressed to the Uniform Classification Committee Agent, G.F. Earl, Tariff Publication Officer, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

NATIONAL BUREAU OF STANDARDS
Handbook H28 - Screw Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

3.1 Qualification. Battery cells and elements furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.3 and 6.5).

3.2 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between requirements of this specification and the specification sheet, the latter shall govern.

3.3 Electrical requirements.

3.3.1 Life. Life, in equivalent cycles or age, shall be as specified in the applicable specification sheet and shall meet the requirements of the guarantee (see 6.4).

3.3.2 Capacity. The capacity of each 126-cell or 252-cell group based on an initial electrolyte temperature of 26.7°C, shall be as specified in the applicable specification sheet. The average and final voltages specified in the applicable specification sheet reflect the voltage drop in the intercell connectors based on a 2 percent loss in the average voltage at the 1-hour discharge rate. The capacity for each of the various discharge rates of the applicable specification sheet is the rated capacity. The battery shall have a minimum of 90 percent of the 6- or 10-hour rated capacity with five cycles after electrolyte filling to demonstrate the initial capacity (see 4.5.3.1 and 6.4). The rated capacity shall be attainable within the number of equivalent cycles specified in the applicable specification sheet.

3.3.3 Cell terminals. The current density of cell terminal bus bar contact areas shall not exceed 39 amperes per square centimeter (cm) at the 1-hour discharge rate.

3.3.3.1 Terminal resistance. The resistance between the terminal copper insert and the lead bus bar shall be less than 13 microohms when tested in accordance with 4.3.5.

3.3.4 Hydrogen evolution. The hydrogen evolution rate, at the time the cell reaches 90 percent of its rated capacity, shall not exceed the value specified in the applicable specification sheet. In addition, the hydrogen evolution rate during the service life routine (see 4.3.4.2.2) shall not exceed the value specified in the applicable specification sheet.

3.4 Physical requirements.

3.4.1 Dimensions. Cell dimensions shall be as specified in the applicable specification sheet.

3.4.2 Mass. In the "filled and charged" condition, the average cell mass and the maximum cell mass shall be within the values shown in the applicable specification sheet.

3.4.3 Cell jars. Unless other specified (see 6.2.1), cell jars shall be in accordance with MIL-J-16120. The jars shall be capable of meeting the dielectric requirements specified in NAVSEA S9086-GI-STM-010/CH-223 V1.

3.4.4 Cell covers. Cell covers shall be in accordance with DOD-P-15816. The covers shall accept a flash arrestor in accordance with MIL-A-22681 so installed in the vent opening that no leakage of gases or electrolyte will occur around the edges. The flash arrester shall be the only cover fitting which will permit the passage of gases from the cell. Service opening(s) equipped with gas-tight removable plug(s) shall be provided for watering and taking readings. Threaded apertures shall be provided on each side of the cover so that connection of the electrolyte circulating tube can be made. Any unused opening shall be sealed with a plug and soft rubber washer.

3.4.5 Cover-to-jar sealing. The cover-to-jar seal shall be battery sealing compound in accordance with MIL-C-2687.

3.4.6 Electrode pack assembly. Positive and negative electrodes shall be formed prior to final assembly into elements.

3.4.6.1 Grids. Grids shall be lead-antimony alloy and shall not exceed 9.0 percent antimony by mass. The grid material shall not contain impurities in quantity higher than as specified in table I. Grids shall be free of open sections, evidence of cold pouring, and blowholes. When lead burning is used, the resulting bond shall be uniform and free of obvious voids, cracks, and other foreign matter.

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3.4.6.2 Active material. Active material of plates shall not have more breaks or loose parts than allowed by the manufacturer's inspection system plan (see 4.1.1). Active material shall not contain impurities in quantity higher than as specified in table I.

TABLE I. Maximum impurity concentration by mass in active materials and grids in parts per million (p/m).

Impurity	Concentration		
	Positive	Negative	Grid
Iron	200 p/m	200 p/m	120 p/m
Copper	50 p/m	50 p/m	500 p/m
Bismuth	1000 p/m	1000 p/m	500 p/m
Calcium	10 p/m	10 p/m	See note 2
Barium	1 p/m	Allowed	80 p/m
Arsenic	1 p/m	1 p/m	500 p/m
Silver	10 p/m	10 p/m	10 p/m
Nickel	1 p/m	1 p/m	5 p/m
Mercury	0.1 p/m	0.1 p/m	0.1 p/m
Platinum	0.1 p/m	0.1 p/m	0.1 p/m
Aluminum	50 p/m	50 p/m	5 p/m
Zinc	10 p/m	10 p/m	20 p/m
Tin	10 p/m	10 p/m	See note 2
Manganese	10 p/m	0.5 p/m	10 p/m
Strontium	10 p/m	10 p/m	50 p/m
Magnesium	50 p/m	50 p/m	50 p/m

NOTES:
1. The sum of all other impurities shall not exceed 50 p/m with no individual impurity exceeding 5 p/m.
2. Tin and calcium are not contaminates for this grade of battery service and vary in amounts dependent upon manufacturers' specifications and processes.

3.4.6.3 Physical characteristics. The electrodes shall fall within the limits of tolerances established at the time of qualification for the applicable type. The physical characteristics of the electrode pack assembly shall be as described for the qualification test samples and shall be submitted at the time of qualification (see 4.3.2). They shall include the following:

(a) Density of active material (after electrochemical formation and drying of the paste).
(b) Electrode dimensions, including thickness.
(c) Mass of active material per electrode and gross mass of electrodes.
(d) Number of positive and negative electrodes.
(e) Separator system description from positive electrode to negative electrode (see 3.4.7).
(f) Separator characteristics in terms of thickness, porosity, average pore size, resistivity, and analysis (see 3.4.7).

3.4.7 Separators. Separators of microporous rubber shall be used. Fiber glass mats and perforated or slotted sheet retainers may be used in combination with microporous rubber separators. Vertical binding strips are required on both positive and negative electrodes. The characteristics of the microporous rubber separator shall be as follows:

(a) Porosity: 50 to 65 percent.
(b) Electrolyte resistivity in sulfuric acid of 1.280 specific gravity: 2.5 to 6.4 ohm-cm.
(c) Moisture content by mass: not to exceed 2 percent in a maximum of 75 percent relative humidity at a maximum temperature of 35°C.
(d) Impurities: not to exceed the following limits by mass:

Aluminum	0.10 percent
Copper	0.003 percent
Iron	0.05 percent
Magnesium	0.05 percent
Manganese	0.001 percent

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Chloride (as Cl): not to exceed 0.007 percent by mass.
Total metallics other than aluminum, iron, and manganese not to exceed 0.073 percent by mass.

3.4.8 Terminals. Terminal parts and bus bars shall be lead-coated copper. Terminal intercell connector bolts, nuts, and washers shall be stainless steel in accordance with ASTM A276, type 316. Bolts shall be in accordance with MIL-S-1222; nuts in accordance with MIL-N-25027, MS 21045, and figure 2; and washers in accordance with FF-W-92. Lead coating on parts not exposed to direct action of electrolyte shall be applied by electroplating and shall be at least 0.05 mm thick. Copper parts that are exposed to the direct action of electrolyte shall be protected by a coating of lead or lead alloy with no less than 6 mm thickness. Terminal posts shall be marked as specified in 3.4.14.

3.4.8.1 Cooling water channels. The cells and elements shall be provided with channels through the terminal posts and straps for the circulation of cooling water in accordance with the applicable specification sheet. Plugs shall be supplied for each channel opening. Each terminal post and strap assembly shall withstand the test of 4.5.4.5 and shall show no leaks. The construction of the channel in the terminal posts and plate straps shall be such that cooling water in passing through the channel shall not be in contact with lead or lead alloys.

3.4.9 Intercell connectors and jumpers. The intercell connectors, which will be provided by the battery installing activity, shall be lead-coated copper with a coating of 0.08 mm thickness. The connector (or jumper), bolts, and nuts shall be tightened to a torque of 57-62 newton-meter (Nm). The cell assembly shall be capable of withstanding this torque without damage. Cell jumpers shall be supplied by the battery manufacturer as a part of the onboard repair parts (see 3.6.4). The length of cell jumper assemblies shall be approximately 0.75 meter (m) and 1.50 m (see table III), in accordance with Drawing S6204-H-3,211,595, of extra flexible rubber covered cable, type SHOF conforming to MIL-C-915 and MIL-C-915/6. Between any two jumpered terminal posts, the jumper assembly shall be not less than 760 square millimeters (mm^2) in area. Terminal adaptors shall be used to interconnect individual cell terminal posts.

3.4.10 Rubber parts. Soft rubber parts shall be in accordance with DOD-P-15817. Hard rubber parts (except jars, separators, sheathings, and baffles) shall conform to DOD-P-15816.

3.4.11 Electrolyte.

3.4.11.1 Electrolyte composition. Sulfuric acid electrolyte shall be in accordance with O-S-801. Electrolyte specific gravity, when fully charged, shall be as specified in the applicable specification sheet.

3.4.11.2 Electrolyte impurities. The amount of impurities in the electrolyte of a fully charged cell shall not exceed the limits specified in table II at the time the battery reaches its initial capacity.

TABLE II. Maximum impurity concentration in electrolyte.

Impurity	Determined as	Maximum concentration
Ammonium	NH_4	200 p/m
Antimony	Sb	25 p/m
Arsenic	As	1 p/m
Chlorine (including Perchlorates)	Cl	20 p/m
Copper	Cu	20 p/m
Iron	Fe	60 p/m
Manganese	Mn	0.6 p/m
Nickel	Ni	0.5 p/m
Nitrates	NO_3	20 p/m
Platinum	Pt	0.1 p/m
Suspended matter	--	Lead compounds only

3.4.11.3 Electrolyte agitator. The cells and elements shall be provided with an electrolyte agitating tube or tubes designed so that an air supply from outside the cell can be brought into the tube approximately 300 mm below the collar of the air supply tube. The minimum pumping rate of 15 liters of electrolyte per hour shall be obtained when air is applied at a rate of 850 milliliters (mL) per minute and electrolyte is at minimum height

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The connection to the electrolyte pumping tube shall be designed to extend through either of two apertures in the cover. The contractor shall furnish gas-tight and acid-tight connections to the cell or element.

3.4.12 Flash arresters. The flash arresters shall be in accordance with MIL-A-22681.

3.4.13 Assembly of cells and elements. Cells and elements (see 6.6.2 and 6.6.3) shall be free of lead droppings or run downs between the plates that could cause cell failure. The cells shall be constructed in a manner to permit the installation or removal of an element without the use of lead burning. Cells shall be so assembled that each can be easily examined internally. The elements shall be inserted in the jars and the covers sealed in accordance with the procedures specified in NAVSEA S9086-G1-STM-010/CH-223 V1. Assembled cells shall withstand an internal air pressure of 7 kilopascals (kPa) for 1 minute without leakage.

3.4.14 Marking of parts. Elements shall be marked with a serial number and the contractor's name. The contractor's name shall not be marked on the cell cover. Hard rubber parts shall be molded with a piece number or letter for identification. Positive terminal posts shall be marked "POS". Negative terminal posts shall be marked "NEG". The terminal connecting strap shall be marked in accordance with 4.3.5.

3.4.15 Mercury. Mercury shall not be used in the manufacture or in any test or test equipment in which there is any possibility of introducing mercury into the battery through accident or otherwise.

3.5 Environmental requirements.

3.5.1 Temperature. Cells or elements shall not be adversely affected by environmental temperature of minus 18°C to 55°C in storage (dry condition) or 0°C to 55°C after installation (wet condition).

3.5.2 Shock. Cells shall conform to the requirements for grade A, class I, type A, medium weight equipment specified in MIL-S-901.

3.6 Deliverable hardware.

3.6.1 Sets of cells or elements. The number of sets as defined in 6.6.1 shall be as specified (see 6.2.1). For cells, each set shall contain 50 percent cells of the right hand configuration and 50 percent cells of the left hand configuration. Cells or elements shall be delivered formed and dry.

3.6.2 Spare cells or elements. When specified (see 6.2.1), six spare cells or elements (see 6.6.4) shall be furnished.

3.6.3 Terminal parts. Cells and elements shall be provided complete with terminal intercell connector nuts, bolts, and washers. Elements shall be provided with 20 percent of the required seal nuts.

3.6.4 Repair parts and tools. When specified (see 6.2.1), repair parts and tools listed in table III shall be furnished with each quantity of 126 cells.

TABLE III. Repair parts and tools.

Quantity		Distribution	
Name of part	Number supplied	Installation and repair parts	Onboard repair parts
Flash arrester assembly	12	2	10
Hard rubber seal nuts	20	10	10
Hard rubber filling plugs	15	5	10
Complete sets of soft rubber gaskets	15	5	10
Electrolyte agitation air supply tube	131	5	126
Special agitation air supply tubes, long	10	--	10
Sets of electrolyte agitation fittings	10	5	5

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TABLE III. Repair parts and tools. - Continued

Quantity		Distribution	
Name of part	Number supplied	Installation and repair parts	Onboard repair parts
Sets of insulated wrenches	2	1	1
Electrolyte height indicators	2	--	2
Cell jumper cable assembly:			
0.75 m	8	--	8
1.50 m	4	--	4

3.6.4.1 Electrolyte height indicator. Electrolyte height indicating devices shall conform with the instructions specified in NAVSEA S9086-G1-STM-010/CH-223 V1.

3.6.4.2 Wrenches. When repair parts and tools are specified (see 3.6.4), steel wrenches coated with acid-resistant electric insulation shall be provided as follows:

(a) Terminal nut socket wrench.
(b) Terminal nut spanner wrench.

3.7 Engineering services (see 6.2.1). The services of the contractor's engineer shall be provided when requested by the Naval Ship Engineering Center (NAVSEC) in order to:

(a) Assist the shipyard in preparing the battery for installation and installing it.
(b) Assist the ship's force in becoming familiar with the battery.
(c) Examine the battery condition and take corrective action in case of unexpected problems.
(d) Participate in technical meetings with NAVSEC or the ship's force when involving the batteries and in locations other than the manufacturer's plant.

An average of 15 mandays will be allowed for the entire life of each battery installed. The maximum number of mandays shall be as specified in the contract or order for the entire number of batteries which can be assembled from cells or elements bought on the same contract. A field engineering report shall be prepared in accordance with the data ordering document included in the contract or order (see 6.2.2).

3.8 Technical data. The contractor shall prepare technical data in accordance with the data ordering documents included in the contract or order (see 6.2.2) and as specified in 3.8.1 and 3.8.2. The contractor shall use the International System of Units (SI) in all technical data. For threaded hardware designation, the Unified Thread Standards of Handbook H28 shall be used.

3.8.1 Drawings. Engineering drawings and a Curves and Data drawing shall be in accordance with the format of Drawing 9-S-5091-L. When such drawings have been submitted, a certificate of prior submittal will be acceptable unless otherwise specified in the contract or order.

3.8.2 Manual. A manual which shall be referred to as a "service manual" shall conform with NAVSEA S9086-G1-STM-010/CH-223 V1, making reference to specific articles of this technical manual, wherever applicable. When an applicable manual has been previously submitted, a certificate of prior submittal will be acceptable unless otherwise specified in the contract or order.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection system. The contractor shall provide and maintain an inspection system acceptable to the Government for supplies and services covered by this specification. The inspection system shall be in accordance with MIL-I-45208 (see 6.2.1).

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Qualification inspection (see 4.3).
(b) Quality conformance inspection (see 4.5).

4.3 Qualification inspection. Qualification inspection shall be conducted at a laboratory satisfactory to NAVSEC. Qualification inspection shall consist of the examination specified in 4.4 and the tests specified in 4.3.1 through 4.3.5.

4.3.1 Curve and data sheet. Prior to the submission of the sample cells specified in 4.2.2, a curve and data sheet, in accordance with the format of Drawing 9-S-5091-L, shall be submitted to NAVSEC for acceptance.

4.3.2 Qualification samples. Four sample cells in the "uncharged and dry" condition of each type for which qualification is desired shall be submitted. One sample shall be subjected to the high-impact shock test of 4.3.3 and three samples shall be subjected to the life test of 4.3.4 and the terminal resistance of 4.3.5. Qualification shall include all classes covered by the applicable specification sheet.

4.3.3 High-impact (H.I.) shock test. A "charged and wet" cell shall be tested to determine conformance with 3.5.2. The cell shall be mounted in a manner duplicating shipboard installation as nearly as possible, and at an angle such that the plane of the cell plates is parallel to a plane tilted down 45 degrees from the vertical. The cell shall be delivering current at a constant rate between the 3-hour and 10-hour rates during testing. Voltage and current shall be continuously recorded for a time interval lasting from not less than 15 seconds before to not less than 1 minute after each shock blow and instantaneously recorded every 5 minutes between shock blows and every 15 minutes after the last shock blow until the cell has discharged to the test end voltage limit for the current rate delivered (see accepted curve and data sheet of 4.3.1).

4.3.4 Service life test.

4.3.4.1 Capacity. The qualification samples shall be first subjected to the test specified in 4.5.4 for cells or elements shipped in the "dry" condition. The test of 4.5.3.5 for demonstration of 100 percent of rated capacity within the number of equivalent cycles specified in the applicable specification sheet shall be met in the course of the life test schedule specified in 4.3.4.2.

4.3.4.2 Life test schedule. The discharge rates and voltages shall be in accordance with the accepted curve and data sheet. Upon successful completion of the capacity test specified in 4.3.4.1, the cells shall be subjected to life testing on the regime specified in 4.3.4.2.1 through 4.3.4.2.10. The cells shall yield not less than the equivalent number of cycles specified and defined in the applicable specification sheet, exclusive of the charging performed during the activation procedure specified by the manufacturer and accepted by NAVSEC.

4.3.4.2.1 Capacity discharges. A series of discharges at the 10, 1, 48 and 1/2 hour rates shall be made. Discharge rates and limiting voltages shall be taken from the accepted curve and data sheet. Initial temperatures shall be 32°C ± 5°C for each discharge. No cooling shall be provided during the capacity discharges.

4.3.4.2.2 Life test routine. The cells shall be subjected to the following life test routine, which will cover 63 days. During the cycling and capacity discharges where a cooling system is provided, cooling water shall be circulated at a temperature of 35°C ± 5°C through the channels provided in the terminal posts and straps at a rate of 1.5 liters per minute per cell. No cooling shall be provided during the hydrogen evolution test. For cells equipped with electrolyte agitators, agitators shall be operated during recharges by supplying 850 mL of air per minute per cell. Operation of the agitator shall be continued for at least 1 hour after the completion of each charge (see 4.3.4.2.3).

(a) Each day for 42 days, cells shall be discharged for 18 hours at an average routine discharge rate as specified in the applicable specification sheet, followed by a modified constant potential charge until the normal charge requirements are met. The remainder of the 24-hour period shall be on open-circuit. On alternate days, a discharge shall be interposed at the 1-hour rate for 15 minutes or to prior final voltage. The partial 1-hour rate discharge shall be preceded and followed by the routine discharges to make up a total of 18 hours. The preceding discharges shall be for zero hours on the first day, 1 hour on the third day, 2 hours on the fifth day and so on up to 17 hours on the thirty-fifth day. On all other days, the discharge shall be 18 hours at the routine discharge rate.

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(b) Every fifth day or every seventh day as specified by the manufacturer, the normal charge shall be extended for 1 hour. This extension shall be applied up to the first 41 days only.
(c) On the 42nd day, the charge shall be made an equalizing charge. After 24 hours on open circuit, the equalizing charge shall be repeated.
(d) Holding the temperature at 38°C + 1°C, measure the hydrogen evolved over the 13 hours following completion of the equalizing charge of item (c).
(e) Discharge the cells at the 10-hour rate to the average final cell voltage or the minimum final cell voltage. Follow the discharge by an equalizing charge.
(f) For the remainder of the 63 days, the cells shall be on open circuit except that on the 63rd day, an equalizing charge shall be given. A new routine shall be started.
(g) On alternate routines run the series of discharges of 4.3.4.2.1.

4.3.4.2.3 The life test routine of 63 days shall be repeated until the capacity at any rate falls below 80 percent.

4.3.4.2.4 Cells shall be watered whenever necessary to prevent the electrolyte level from falling below the minimum specified on the accepted curve and data sheet. In addition, the cells shall be watered prior to each equalizing charge. Cells shall not be watered above the maximum specified on the accepted curve and data sheet.

4.3.4.2.5 For each charge, the following data shall be recorded for each cell:

(a) Specific gravity and temperature at the start and end of the charge.
(b) Voltage and finishing rate just prior to the end of the charge.
(c) Hours of charge to the nearest tenth.
(d) Minutes on charge above the temperature voltage gassing (TVG) point at the finishing rate.
(e) Ampere-hours of charge.
(f) Kilowatt hours of charge.
(g) Quantity of water used, if any.

4.3.4.2.6 For each equalizing charge, in addition to the data of 4.3.4.2.5, the specific gravity corrected for temperature and height of electrolyte 12 hours after completion of the charge shall be recorded. The cells shall be on open-circuit during this phase of testing.

4.3.4.2.7 For each hydrogen evolution test, rate of hydrogen evolution in cubic centimeters per hour per 1000 ampere-hour capacity at the 10-hour rating, reduced to 0°C and one atmosphere pressure, for each hour of stand shall be recorded. The temperature shall also be recorded for each hour of stand.

4.3.4.2.8 For representative discharges and the following recharges the following shall be recorded at sufficient intervals to enable a smooth curve to be plotted against time:

(a) Specific gravity and voltage.
(b) Electrolyte temperature.
(c) Inlet and outlet cooling water temperatures.
(d) Ambient temperature.
(e) Cooling rate in kilocalories per minute.
(f) Cooling water flow rate in liters per minute.

4.3.4.2.9 Progress reports shall be prepared and forwarded to NAVSEC as soon as possible. Each report shall cover the results of two 63-day routines. The reports shall include the following:

(a) Corrected specific gravity and voltage at end of each equalizing charge.
(b) Corrected specific gravity 12 hours after the end of each equalizing charge.
(c) Finishing rate of charge.
(d) Hydrogen evolution in cubic centimeters per hour per 1000 ampere-hour at the 10-hour rating reduced to 0°C and one atmosphere pressure, for the last 6 hours of the 13-hour stand.
(e) Capacity and thermal data for test discharges. The final voltage for each cell shall be given.

4.3.4.2.10 Following completion of the test, a final report shall be submitted. Photographs of the plates showing the condition of the grids and the active material, interval short-circuits and other pertinent features shall be included.

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4.3.5 Terminal resistance. The contact resistance between the terminal copper insert and the lead bus bar shall be measured in accordance with the procedure of NAVSEA S9086-G1-STM-010/CH-223 V1. Terminal connection straps which meet the requirements of 3.3.3.1 shall be stamped with the letter "O". The values obtained shall not exceed the requirement specified in 3.3.3.1.

4.4 Visual examination. Parts and assembled batteries shall be visually examined to verify conformance with all of the requirements of this specification which do not involve tests.

4.5 Quality conformance tests.

4.5.1 Objective quality evidence for material. Objective quality evidence shall be provided as to conformance with requirements specified in section 3 herein. The evidence shall document actual inspection or other verifiable quality data.

4.5.2 Test conditions. The test cycles of "charge" and "discharge", having once begun, shall succeed one another as rapidly as temperature limitations will permit. The temperature of the electrolyte shall be as near 26°C as possible at the start of each discharge. If the temperature is not 26°C, a temperature correction shall be made on the length of the discharge as specified on the accepted curve and data sheet. Charges shall conform as closely as possible to the accepted curve and data sheet. Before test, the battery may be given not more than five cycles for the purpose of developing the capacity.

4.5.3 Tests of cells shipped "filled and charged".

4.5.3.1 Initial capacity test of cells at 10-hour or 6-hour rate. In addition to the general tests specified in 4.3.2 each set of cells to be shipped "filled and charged" shall be tested as follows: Cells shall be assembled and connected in groups convenient to the manufacturer. Each such group shall be discharged at the 10-hour rate or 6-hour rate. Ninety percent of rated capacity shall be obtained, including any correction for temperature, before either the limiting average cell voltage or minimum cell voltage specified on the accepted curve and data sheet is reached. As many as five cycles of charge and discharge will be permitted to develop the capacity, any cells failing this test shall be cause for rejection. If it is definitely shown that the failure is due to mechanical defects such as a cracked jar, and broken separators, rejected cells may be resubmitted for test at a later date. Cells which require the entire five cycles of charge and discharge to pass the test at the 10-hour rate or 6-hour rate shall not be given any additional working at this rate; however, if the test is passed in less than five cycles, the five cycles, but no more, may be completed at the contractor's discretion prior to the 1-hour rate discharge test.

4.5.3.2 Test of electrolyte after discharge. At the end of the discharge test of 4.4.3.1, three cells shall be selected and a sample of 1-liter of electrolyte drawn from each of them. The three samples of electrolyte shall be kept separate and shall be analyzed by the manufacturer to ascertain that the amounts of impurities present do not exceed the limits specified in table II (see 3.4.11.1). If any sample exceeds the limits, the entire group of cells represented by the three samples of electrolyte shall be rejected. Rejected groups of cells may not be offered again for delivery unless a procedure for correction of the deviation has been accepted by NAVSEA, and when resubmitted, the group of cells shall be re-tested at the 10-hour or 6-hour rate and again sampled to determine the amount of impurities present in the electrolyte.

4.5.3.3 Discharge test of six-sample cells at 1-hour rate. The lowest six cells which have passed the initial 10-hour or 6-hour rate discharge test specified in 4.5.3.1 shall, after recharge, be discharged at the 1-hour rate. Ninety percent of the rated capacity shall be obtained, including temperature correction. At the end of the discharge, the average cell voltage shall be not less than that specified on the accepted curve and data sheet, and none of the cells shall be below 1.28 volts. If the average voltage is less than that specified or if any cell is below 1.28 volts, the entire group of cells represented by the sample cells shall be rejected. This test shall be met within six cycles including the test cycles given to meet the 10-hour or 6-hour rate specified in 4.5.3.1.

4.5.3.4 Weight of complete cells. The six cells tested in accordance with 4.5.3.3 shall be weighed individually with the electrolyte at normal level. Cell connectors and terminal bolts shall not be included.

4.5.3.5 Discharge test of three sample cells. Provided the six cells tested as specified in 4.5.3.3 from the first set on each contract or order have been accepted, three of the six cells shall be selected for further tests. The three cells shall be cycled to determine that they will give full service capacity within the number of cycles specified

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and defined on the applicable specification sheet. Full service capacity shall be demonstrated at the 1-hour and 10-hour or 6-hour rates. Service capacity shall be as specified on the accepted curve and data sheet. The cells may be cycled in accordance with the manufacturer's preferred procedure. If the three cells equal or exceed 100 percent of the service capacity within the specified cycles, they will be considered as representing all the batteries of the same type furnished in the contract or order. If they fail to give the required capacity, the group of cells from which they are taken shall be subject to rejection. The 100 percent capacity test shall always be run on the first set on each contract or order unless a 100 percent test has been run on cells or elements of identical type and design within one calendar year prior to the date on which the first set is to be shipped. In that event, the 100 percent tests shall be run on the first sample subjected to the tests of 4.5.3.1 through 4.5.3.3 after the said calendar year has elapsed (see 6.3). At the option of the manufacturer, sets of batteries or elements may be shipped prior to completion of the 100 percent capacity test. However, if upon completion of this test, the 100 percent capacity has not been met, batteries or elements already shipped shall be subject to rejection.

4.5.3.6 Hydrogen evolution test of three sample cells. Upon completion of the test specified in 4.5.3.5, the three cells shall be further tested to determine the rate of hydrogen evolution. Following an equalizing charge, the cells shall be on open circuit for 13 hours. During this time the cell temperature shall be held at 38°C + 1°C. The rate of hydrogen evolution shall be measured at hourly intervals beginning 7 hours after the charge is complete. The average rate of evolution shall be expressed as cubic centimeters per hour per cell per 1000 ampere-hours at the 10-hour rate, reduced to 0°C and one atmosphere pressure. If the average rate of emission of hydrogen exceeds that specified on the applicable specification sheet, the set of cells or elements from which samples were taken shall be rejected. Rejected cells shall not be offered again unless a procedure for correcting the excessive emission of hydrogen has been accepted by NAVSEC, and when resubmitted, the entire group of cells or selected sample cells shall be subjected to such tests as NAVSEC may direct.

4.5.3.7 Disposition of test samples. The three sample cells which have been tested in accordance with 4.5.3.5 and 4.5.3.6 shall be disposed of as follows:

(a) If the cells demonstrated 100 percent rated capacity in five cycles or less (other than those allowed in 4.5.3.1 and 4.5.3.3) they may be shipped in the "charged and wet" condition as part of the set which they represented.
(b) If the cells required more than five cycles but not more than ten cycles to reach 100 percent capacity, they may be shipped in "uncharged and dry" condition or scrapped at the discretion of the manufacturer with the concurrence of the Government inspector. If shipped "uncharged and dry" they will be accepted as repair part cells on the contract or order, and three other cells shall be furnished in the "charged and wet" condition to replace the test cells.
(c) If the cells required more than ten cycles to develop 100 percent capacity, they will not be accepted as a delivery on the contract or order. Three other cells shall be furnished in the "charged and wet" condition to replace the test cells.

4.5.3.8 Shipping condition, filled cells. Cells shall be shipped filled and fully charged. Tested batteries and cells shall be recharged and the specific gravity of the electrolyte in each cell shall be adjusted after charging so that the difference between any individual cell gravity and the average for the group of cells shall be not more than plus or minus 0.005 with electrolyte at normal level. An equalizing charge shall be given within 15 working days of the date of shipment.

4.5.4 Tests of cells or elements shipped in dry condition.

4.5.4.1 Sample cells or elements shall be selected for tests from the first set of cells or elements and from every third set thereafter (4th set, 7th set, 10th set and so forth) except that when the sampling interval results in selection of a sample from the last set on a contract or order, the manufacturer may elect that the sample be taken from the next to the last set, instead. The number selected shall be approximately 2-1/2 percent of the number in the set sampled, but never less than three cells or elements. Sample elements shall be set up for test in standard containers.

4.5.4.2 Tests of samples selected in accordance with 4.5.4.1 shall be conducted as specified in 4.5.3.1 through 4.5.3.3 and all of the technical provisions of the manufacturer's test plan for cells shipped in the filled condition. The manufacturer's instructions for putting the battery or set or elements into service shall be accepted by NAVSEC. Three cells or elements shall be tested as specified in 4.5.3.5 and 4.5.3.6 after the tests of 4.5.3.1 through 4.5.3.3. Any test cells or elements which fail to pass the required tests

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shall be rejected. If two or more test cells or elements fail to pass any of the required tests, the set represented by the sample shall be rejected. The rejected set of cells or elements, however, may be reconditioned and resubmitted as a lot for retest at a later date, subject to acceptance of NAVSEC. Shipment may be authorized by the Government inspector upon completion of the 10-hour or 6-hour and 1-hour test discharges. Test cells or elements may be bone-dried and delivered as spare cells or elements.

4.5.4.3 Chemical analysis after tests. A sample of active material shall be selected from the positive and negative plates of test cells redried for use as spares to determine that the degree of moisture and free acid present in the active material shall not exceed 0.5 percent for moisture or 0.1 percent for free acid.

4.5.4.4 Shipping condition, dry cells or elements. Test cells or elements tested in accordance with 4.5.3.5 may be placed in the "uncharged and dry" condition and shipped as spare cells or elements provided that not more than ten cycles have been given. At the discretion of the manufacturer the tested cells may be replaced by other cells or elements which have been tested in accordance with 4.5.3.1. If more than ten cycles have been given, the cells or elements shall be scrapped and replaced.

4.5.4.5 Terminal posts and strap assembly. Sampling for terminal posts and strap assemblies shall be in accordance with MIL-STD-105 and shall have an AQL of 1 percent. The sample(s) terminal post and strap assembly shall be tested under water with not less than 210 kPa air pressure to determine conformance with 3.4.8.1.

4.6 Laboratory test cells. When specified (see 6.2.1), the contractor shall also furnish three sample cells of the same type for each type cell or element furnished in accordance with this specification and deliver to the Chemical Laboratory, Mare Island Naval Shipyard, Vallejo, California. These shall be selected at random for tests to investigate the characteristics of the particular types of batteries, so that proper instructions may be prepared for the maintenance of these batteries in service. Whenever the tests performed are simulated service life tests, the results of such tests may be considered by the Government in any determination leading to a claim by the Government against the contractor as a result of the contractor's liability under the provisions of the guarantee cited in section 6 of this specification. The cells shall be shipped in the bone-dry condition, and will be filled and charged at the laboratory. Instructions for the initial filling and charging, and procedures for conducting freshening, normal and equalizing charge shall be furnished with the cells.

4.7 Inspection of preparation for delivery. The cleanliness, preservation-packaging, packing, and marking shall be inspected for compliance with section 5 of this document.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government acquisitions. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.8.)

5.1 Cleanliness, preservation-packing, packing, and marking. Cleanliness, preservation-packing, packing, and marking shall be level A as specified herein.

5.2 Cleanliness (level A). Cleanliness of materials used for packaging and packing shall be sufficient to prevent the introduction of impurities (see table I) into cells or elements.

5.3 Preservation-packaging, and packing (level A).

5.3.1 Cells and elements. Cells and elements shall be assembled for shipment in packing containers accepted by NAVSEC. Containers shall be of the overseas type or class and designed to handle each cell in the "filled and charged" or "uncharged and dry" condition. Elements shall be supported and secured in a manner that will prevent damage to element and components thereof. Plates, separators, and components shall be secured in a manner that will prevent plate spread movement of separators or other components. Cells and elements shall be packaged and loaded with the plates perpendicular to the direction of motion.

5.3.1.1 The following information shall be sealed in a waterproof bag or envelope and packed in the first packing container and each additional fiftieth packing container:

(a) A copy of the accepted Curves and Data drawing shall be furnished for cells in the "filled and charged" condition.

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(b) A copy of the accepted Curves and Data drawing and approved instructions for putting into service shall be furnished for cells or elements in the "dry and uncharged" condition.

5.3.2 Spare cells and elements. Spare cells and elements shall be packed as specified in 5.3.1. The tops of packing containers containing spare cells or elements shall be painted gray. A copy of the approved instructions for putting "uncharged and dry" cells and elements into service and a copy of the accepted Curves and Data drawing shall be packed with each spare cell or element.

5.3.3 Installation repair parts and tools. A set of installation repair parts and tools (see table III) shall be packed in a single shipping container in a manner that will insure arrival at the destination in satisfactory condition and which will be acceptable to the carrier at the lowest rates. Containers and packing shall comply with Uniform Freight Classification Rules or other carrier regulations as applicable to the mode of transportation. A list of contents shall be packed within the shipping container.

5.3.4 Onboard repair parts and tools. A set of onboard repair parts and tools shall be packed in the same manner as specified in 5.3.3.

5.3.5 Manuals. Technical manuals furnished with a set of cells and elements shall be packaged in a transparent waterproof plastic bag, minimum 4 mil thick. Bag closure shall be by heat sealing. The packaged manuals shall be placed in the number one container of the shipment (see 5.4(k)). Bulk manuals shall be prepared for shipment in accordance with the level specified of MIL-M-15071 (see 6.2.1).

5.4 Marking (level A). Packing cases for cells and elements shall be marked in accordance with MIL-STD-129. In addition each packing case shall be clearly marked and numbered to facilitate ready identification regardless of stowage position. Markings on the exterior of packing cases shall include:

(a) Submarine main storage batteries.
(b) Contractor's name.
(c) Contract number.
(d) Item and set number, as applicable.
(e) Box serial number with total number of boxes comprising the set or item as applicable.
(f) A brief identifying description of the material in the containers such as "Cells, Main Storage Battery" "Dry and uncharged" or "Wet and charged", "Spare cell" or "Spare element".
(g) Contractor's type designation.
(h) National stock number (furnished by NAVSEC upon request).
(i) Mass and cubage.
(j) FRAGILE, HANDLE WITH CARE. THIS END UP.
(k) Mark crate number 1 as follows: "CONTAINS MANUALS" (see 5.3.5).

5.4.1 Installation repair parts and tools. Shipping containers shall be marked for shipment in accordance with MIL-STD-129. In addition, the marking shall include "SUBMARINE MAIN STORAGE BATTERY INSTALLATION REPAIR PARTS" and submarine name and hull number, if intended for an individual submarine.

5.4.2 Onboard repair parts and tools. Shipping containers shall be marked for shipment in accordance with MIL-STD-129. In addition, the marking shall state "SUBMARINE MAIN STORAGE BATTERY ONBOARD REPAIR PARTS".

5.5 Technical data (see 3.8). When specified (see 6.2.1 and 6.2.2), packaging-packing drawings shall be submitted. These drawings are not required when previously submitted and accepted.

6. NOTES

6.1 Intended use. This specification covers lead-acid battery cells and elements using lead-antimony alloy grids for use on diesel-electric submarines.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification and applicable specification sheet.
(b) Type and class required (see 1.2).

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(c) Cell jar material if other than as specified (see 3.4.3).
(d) Number of sets of cells or elements (see 3.6.1).
(e) Number of spare cells or elements (see 3.6.2).
(f) Repair parts and tools required (see 3.6.4):
    (1) Onboard.
    (2) Installation.
(g) If engineering services are required (see 3.7).
(h) Quality assurance requirements (see 4.1.1).
(i) Whether sample cells are required (see 4.5).
(j) Level of packing required for bulk manuals (see 5.3.5).
(k) When packaging technical data is required (see 5.5).

6.2.2 Data requirements. When this specification is used in a contract which invokes the provision of the "Requirements for Data" of the Defense Acquisition Regulation (DAR), the data identified below, which are required to be developed by the contractor, as specified on an approved Data Item Description (DD Form 1664), and which are required to be delivered to the Government, should be selected and specified on the approved Contract Data Requirement List (DD Form 1423) and incorporated in the contract. When the provisions of the "Requirements for Data" of the DAR are not invoked in a contract, the data required to be developed by the contractor and required to be delivered to the Government should be selected from the list below and specified in the contract.

Paragraph	Data requirements	Applicable DID	Option
3.7	Report, contractor engineering and technical services (CETS)	UDI-E-23127	---
3.8.1	Drawings, engineering and associated lists	DI-E-7031	Level 3 (production) Design activity designation-contractor Drawing number-contractor
3.8.2	Manual, technical standard	DI-M-2044	MIL-M-15071, type I
5.5	Drawings, preservation-packaging, packing and marking (PPP and M)	UDI-E-23123	---
5.5	Packaging and transportation support data	UDI-P-23509	---

(Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.)

6.2.2.1 The data requirements of 6.2.2 and any task in section 3, 4, or 5 of the specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 Bids. The provisions of 4.5.3 for frequency of performance of the service rating capacity test are such that most large contracts will require performance of this test at some time during the contract. In order to provide a consistent basis for comparison of bids, bids should be based on the assumption that one service rating capacity test per type will be required. If deliveries are required to extend over a period longer than 12 months, it should be assumed that two service rating capacity tests will be required.

6.4 Guarantee. The guarantee shall be as specified in 6.4.1 through 6.4.5.

6.4.1 The contractor guarantees the life of the cells and elements furnished on this contract in terms of time under service operating conditions. Cells or elements furnished shall be guaranteed for the months or equivalent cycles specified in the applicable specification sheet.

6.4.2 The period of guarantee for cells or elements shall be considered to begin on the date the battery is put into service (start of initial charge), except when the time between the date of shipment and the date of initial charging exceeds 2 years, in which case the

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guarantee shall begin 2 years from the date of shipment. The life of the cells or elements shall be considered at an end when the cells cannot be made to deliver 80 percent of their rated capacity, when handled and maintained in accordance with the methods and procedures specified by the contractor's service manual. Defective cells may be jumpered out of the battery so long as the remaining cells are capable of delivering 80 percent of the rated capacity (including any correction for temperature), before either the total battery voltage or any individual cell voltage falls below its cut-off voltage, at the corresponding discharge rate, in accordance with the applicable specification sheet. In case of failure to meet the guaranteed life, it is understood and agreed that the battery manufacturer will indemnify the Government on a prorata basis (proportionately) for equivalent cycles or months of service not obtained. (If P is the purchase price of the cells or elements, n the number of months not obtained, Q the number of equivalent cycles not obtained, m the months of guaranteed service life in the applicable specification sheet, and E the number of guaranteed equivalent cycles in the applicable specification sheet, the smaller of the two numbers $P \times {}^{n}/_{m}$ or $P \times {}^{Q}/_{E}$ shall be used).

6.4.3 At the option of the Government, the amount to be paid the Government may be assessed as a cash indemnity, or may be applied to reduce the cost of any other similar submarine main storage battery or set of elements which the Government may contract to purchase from the contractor. It is understood and agreed that a cash indemnity will only be assessed when, in the opinion of the Government, experience indicates that the type of cell manufactured under this contract by the contractor is unsuitable for use as submarine main storage batteries.

6.4.4 With respect to sets of cells or elements from which no sample was taken for factory acceptance tests prior to shipment, or with respect to sets which the Government may elect to accept even though samples were tested which failed, the manufacturer further guarantees that each such set furnished will, when filled and given an initial charge by the Government or its agent in accordance with manufacturer's instructions, yield ninety percent of the rated capacity at the 10-hour (or 6-hour) discharge rate within the first five cycles (exclusive of those allowed for the development of the initial charge). For this purpose, a cycle consists of a discharge at one or more rates from a fully charged condition to the low voltage limit for the rate of discharge last used, together with the following charge which brings the battery back to the condition which existed at the beginning of the discharge. In the event the specified initial capacity is not obtained within the said five cycles, the manufacturer agrees to pay the Government the amount equal to all reasonable costs incurred by the Government for cycling beyond five cycles as necessary to develop the specified initial capacity. The manufacturer further agrees that use of premium labor, services or materials for such additional cycling will not be the basis for dispute of the amount of said costs, if such is either the normal practice of the activity performing the work or is required to expedite preparation of the battery so as to not interfere with other work.

6.4.5 It is understood and agreed that the above guarantee shall not be construed as to cover normal replacements of separators and jars.

6.5 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening for bids, qualified for inclusion in applicable Qualified Products List QPL-24200 whether or not such products have actually been so listed by that date. The attention of the contractors is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Ship Engineering Center, SEC 6124, Department of the Navy, Washington, DC 20362, and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.5.1).

6.5.1 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

6.6 Definitions. The definitions specified in 6.6.1 through 6.6.5 are applicable to this specification:

6.6.1 Set. A set is the total number of cells or a total number of elements sufficient to assemble one complete battery installation in a ship.

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6.6.2 Element. An element is a complete assembly consisting of the positive group and the negative group of electrode assemblies mounted together with separators and retainers in place, in all respects ready for installation in the jar.

6.6.3 Cell. A cell is an element installed in a jar with cover in place. There are two types of cell configurations; right and left hand. The right hand configuration is shown in the plan view of the cell cover in the applicable specification sheet. The left hand configuration interchanges the position of the service tube and the cell marker.

6.6.4 Spares. Spare cells or elements are cells or elements supplied in excess of the cells constituting a set.

6.6.5 Group. A battery group is 126 cells for a class 2 (see 1.2.2) installation or 252 cells for a class 3 (see 1.2.2) installation.

6.7 Cross-reference of classification. This specification deletes grades LA and LC, and class 1 and 4 of MIL-B-24200A(SHIPS) dated 20 April 1970.

6.8 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.9 Changes from previous issue. The symbol "#" is not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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